Haematology Analyser Upgrade / Business Case

Haven Paediatric Centre / 5-part analyser acquisition, current 3-part relegated to backup

The decision

Buy a Zybio 5-part haematology analyser, promote it to primary, and drop the current 3-part machine to backup. Quoted at $6,316 landed on freight. This is a low-regret capital purchase, but the case for it is redundancy and clinical quality, not new revenue, so it must be justified honestly.

The numbers that matter

A haematology analyser produces the FBC. Malaria (MP) is microscopy or RDT, so it is set aside here. The relevant revenue line is FBC alone (Jan to Jun 2026 actuals from the Head of Ops):

Line	Monthly average	Annualised run-rate
FBC (the analyser's output)	N695,296	N8.34M
All labs (context)	N2,179,576	N26.15M
Machine, quoted landed ($6,316 at ~N1,450/$)	one-off	~N9.2M

Capex is roughly 1.5 months of total lab revenue. The trap: FBC income already exists on the 3-part machine, so a second analyser does not earn it again. Any payback sold against FBC revenue is double-counting.

The real case: three levers

1. Business continuity. A single analyser is a single point of failure. If it dies, N695K/month of FBC stops and children get sent out. A ~N9-13M machine is cheap insurance on an N8.3M/yr line and a N26M/yr lab, and on clinical safety in a paediatric and NICU setting.

2. Clinical quality. 5-part gives the full white-cell differential (neutrophils, lymphocytes, monocytes, eosinophils, basophils) versus the 3-part's coarse split. For neonatal sepsis and paediatric differentials this is a genuine standard-of-care step up.

3. Optionality. Fewer send-outs for 5-part differentials, and a defensible basis to hold or raise the N11,500 private tariff.

Cost reality check

The $6,316 is machine plus freight only. It is not the landed cost. Fully landed, with duty, VAT, clearing, install and training, realistically lands nearer **N11-13M**. Three items decide the call, and none are on the quote yet:

- **Reagents.** The real long-run cost. 5-part reagents are pricier and often proprietary. Confirm cost per test and a local supplier. Reagent lock-in on an orphan machine is the true risk, not the sticker.
- **Service and warranty in-country.** Zybio is a budget brand. A 5-part with no local engineer is worse than the 3-part you trust. Confirm who services it and the warranty terms.
- **Fully landed cost.** Duty, VAT, clearing, installation and training added to the $6,316.

Verdict: Conditional GO. The capex is small, the continuity and clinical-quality logic is sound for a paediatric centre, and the downside is bounded. Approve subject to the three answers above. If reagent cost per FBC sits comfortably under the tariff and local service exists, it is a clear yes. If reagents are proprietary and imported with no local support, pause and price a mainstream brand (Sysmex or Mindray) even at a higher sticker, because uptime beats unit price here.